

Ameya Anjarlekar

✉ ameyasa2@illinois.edu
📄 <https://ameyanjarlekar.github.io/>
🌐 ameyanjarlekar in ameya-anjarlekar-508371166

Education

- 2022–present **PhD, Electrical & Computer Engineering**, *University of Illinois*, Urbana-Champaign, GPA 3.97/4.
Privacy and Fairness, Mechanism Design, Reinforcement Learning, Optimization and Control Theory
- 2017–2021 : **Bachelor of Technology, Electrical Engineering (with a minor in Computer Science)**, *Indian Institute of Technology*, Bombay, CPI 9.64/10.

Key Publications

- NeurIPS **Scalable Policy-Based RL Algorithms for POMDPs**, *NeurIPS 2025*.
- IEEE CDC (Invited talk) **Online Mechanism Design for Differentially Private Data Acquisition**, *IEEE Conference on Decision and Control 2024*.
- AAAI **Efficient LLM Unlearning via Gradient Ratio-Based Influence Estimation and Noise Injection**, *AAAI 2026 (Under Review)*.
- CSSP **Generalized Fractional Ambiguity Function and Its Applications**, *Circuits Systems & Signal Processing, Springer 2020*.

Industrial Experience

- 02/25-08/25 **NVIDIA**, *Responsible AI Intern*.
- Designed a comprehensive evaluation suite by curating and integrating open-source tools and benchmarks to assess **Trustworthy AI** metrics—such as privacy, robustness, bias, and broader safety risks (e.g., biosecurity, cyber threats, loss of control, AI R&D misuse)—across all modalities, including **multimodal** and **agentic LLMs**.
 - Proposed and developed a noise-injected selective **LLM fine-tuning** algorithm to enable targeted machine unlearning by **optimizing** a loss function that penalizes unsafe responses while preserving model utility through a regularization term, aimed at mitigating unsafe responses and reducing potential copyright violations.
 - The work on **machine unlearning** resulted in a paper, and a provisional **patent** has also been filed.
- 06/21-06/22 **Quadeye Securities**, *Quantitative Researcher*.
- Developed arbitrage-based strategies to formulate high-frequency trading algorithms in the derivatives segment
 - Leveraged Machine Learning-based analysis to improve existing strategy by elevating Sharpe ratio from 3 to 4
- 06/20-07/20 **Daikin Industries**, *Machine Learning Research Scientist Intern*.
- Accomplished 50% **video data compression** by developing a Hitomi Camera-inspired image processing algorithm
 - Modified the algorithm by incorporating a data-driven approach to further improve compression performance

Research Experience

- 08/24–present **Scalable Policy-Based RL Algorithms for POMDPs**.
- Proposed an approximate belief state mapping that converts a finite state discounted/average reward **POMDP** into a finite state fully observed MDP with provable approximation error bounds
 - Obtained theoretical bounds demonstrating the error to decay exponentially with the size of the history
- 08/22-08/24 **Mechanism Design for Heterogeneous Differentially Private Data Acquisition**.
- Introduced a mechanism design approach using ideas from game theory and statistical learning to design a data market to model and optimize utility-privacy tradeoff in differentially private training of machine learning models.
 - Developed an asymptotically optimal algorithm for the online version of the problem wherein the data providers approach the platform sequentially and demand instantaneous payments.
- 08/18-12/19 **IIT Bombay Mars Rover Team**.
- Formulated an object detection AI framework using Python essential for the autonomous operations of the Mars rover robot by developing a **computer vision** algorithm using transfer learning, resulting in 98% accuracy
 - Designed and implemented an end to end system using **ROS** and Bash scripts to collect visual data from **LiDAR** and RGB cameras, enabling decision-making for the robotic traversal through the deep learning model.

Technical Skills and Interests

- Skills Statistical Learning, Data Structures, Data Science, Artificial Intelligence, Pattern Recognition, Statistics
- Tools Python, C, C++, ROS, PyTorch, Pandas, Tensorflow, Bash, Git, CI/CD, Matlab, Kubernetes, Slurm
- Interests Privacy and Fairness, Game Theory, Deep Learning, Reinforcement Learning, Computer Vision, Robotics

Awards and Achievements

- * Awarded **Undergraduate Research Award (URA-01)** by IIT Bombay for innovative research contribution
- * Secured an All-India Rank **132** in **JEE - Advanced** and an All-India Rank **215** in **JEE - Main**